

```
In [2]: import customized_sigmoid
        from fit import*
        from data_handler import *
        import os
        from optimizer import*
```

```
In [3]: import matplotlib.pyplot as plt
```

```
In [4]: import torch.optim
```

```
In [5]: import utils
```

```
In [6]: from tqdm import tqdm as tq
        from tqdm import trange
        import time
```

```
In [7]: import cbs
        import unet
```

Here we re-test the new Dataset Anomaly Flood Recovery

Prcedure goes as following:

- 1- Load Old Models
- 2- Load Old results on both validation and anomaly dataset
- 3- Evaluate the new dataset on the old Model
- 4- Gragh and compare results

Why this Experiment is important?

The Importance of this experiment lies in validating the uncertainty we have about test dataset content as the dataset has normal and anomaly result this may weaken our ability to judge performance of the model

What would a great model looks like in graphs ?

A great model will have low validation loss and low loss **anomaly recovery dataset** and relatively high loss for test set

This means that the overlapping occurs between normal and anomaly data is due to model succesfully evaluate normal packets in anomaly dataset as Normal

Data Preparation

```
In [16]: os.listdir('')
```

```
Out[16]: ['Adam', 'val.npy', 'U-net_sigmoid', 'can.npy']
```

```
In [12]: ## Data preparation

test_can , _ = split('dataset/anml_flod_rec/CAN/',prc = 0 , dataset_pct= 0.3, dim = (112,112), word_type=torch.float)
test_eth_short , _ = split('dataset/anml_flod_rec/ETH_short/',prc = 0 , dataset_pct= 0.3, dim = (112,112), word_type=torch.float)
test_can_dl = DataLoader( test_can, 16, num_workers=4,shuffle=False)
test_eth_short_dl = DataLoader( test_eth_short, 16, num_workers=4,shuffle=False)
```

Experiment CAN

Load old results on test

```
In [20]: path = 'Models/Unet_2_epoch_CAN_only_FULL/'
```

```
In [39]: val = np.load(path+'val.npy')
anml_can = np.load(path+'can.npy')
```

Model

```
In [28]: m = unet.Unet_sigmoid().cuda()
```

```
In [29]: opt = torch.optim.Adam(m.parameters(),1e-4)
```

```
In [30]: opt = BaseOptimizer(opt, 'Adam')
```

```
In [31]: learn = learner()
```

```
In [32]: test_item = Validate(test_can_dl)
```

```
In [33]: utils.load_all('Unet_2_epoch_CAN_only_FULL/',learn = learn , model = m , opt = opt)
```

```
In [38]: test_item.evaluate(model = m , crit = torch.nn.functional.binary_cross_entropy)
```

```
0%|          | 0/1358 [00:00<?, ?it/s]
```

```
testing...
```

```
/mnt/5e85af0b-fa6b-4dca-a539-449bff7060b4/Work/logsagenttechnicaldepth/fit.py:144: UserWarning: Using a target size (torch.Size([16, 112, 112])) that is different to the input size (torch.Size([16, 1, 112, 112])) is deprecated. Please ensure they have the same size.
```

```
loss = crit(out,data)
```

```
100%|██████████| 1357/1358 [01:26<00:00, 21.11it/s]/mnt/5e85af0b-fa6b-4dca-a539-449bff7060b4/Work/logsagenttechnicaldepth/fit.py:144: UserWarning: Using a target size (torch.Size([7, 112, 112])) that is different to the input size (torch.Size([7, 1, 112, 112])) is deprecated. Please ensure they have the same size.
```

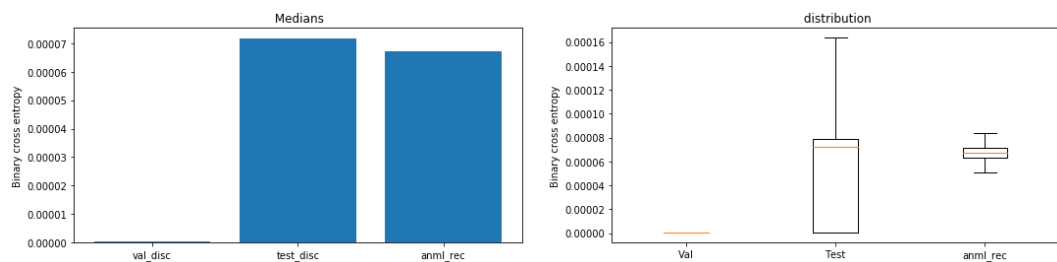
```
loss = crit(out,data)
```

```
In [42]: anml_rec_gen = torch.tensor(test_item.test_loss)
         np.save(path+'anml_rec_gen.npy',anml_rec_gen)
```

```
In [62]: fig , ax = plt.subplots(1,2,figsize=(18,4))
         ax[0].bar(['val_disc', 'test_disc','anml_rec'],[np.median(val),np.median(anml_rec),np.median(anml_rec_gen)])
         ax[0].set_title(f' Medians')
         ax[0].set_ylabel('Binary cross entropy')

         ax[1].boxplot([val,list(anml_rec),list(anml_rec_gen)],showfliers = False);
         ax[1].set_xticklabels(['Val','Test','anml_rec']);
         ax[1].set_title(f' distribution')
         ax[1].set_ylabel('Binary cross entropy')
```

```
Out[62]: Text(0, 0.5, 'Binary cross entropy')
```



Weak results due model in this case will consider all CAN packets from both Anomaly and Recovered state as anomaly

Experiment Short Ethernet

Load old results on test

```
In [44]: os.listdir(path)
```

```
Out[44]: ['Adam', 'eth_short.npy', 'val.npy', 'U-net_sigmoid']
```

```
In [43]: path = 'Models/Unet_2_epoch_ETH_short_only_FULL/'
```

```
In [55]: val = np.load(path+'val.npy')
         anml_eth_short = np.load(path+'eth_short.npy')
```

Model

```
In [46]: m = unet.Unet_sigmoid().cuda()
```

```
In [47]: opt = torch.optim.Adam(m.parameters(),1e-4)
```

```
In [48]: opt = BaseOptimizer(opt,'Adam')
```

```
In [50]: learn = learner()
```

```
In [51]: test_item = Validate(test_eth_short_dl)
```

```
In [52]: utils.load_all('Unet_2_epoch_ETH_short_only_FULL/', learn = learn , model = m
, opt = opt)
```

```
In [53]: test_item.evaluate(model = m , crit = torch.nn.functional.binary_cross_entro
py)
```

```
0%|          | 0/1204 [00:00<?, ?it/s]

testing...

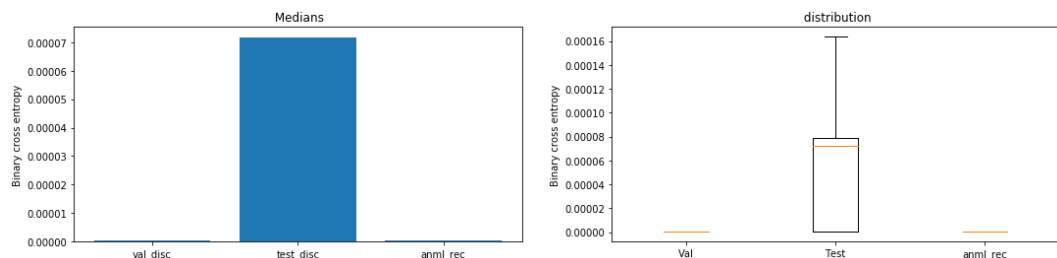
/mnt/5e85af0b-fa6b-4dca-a539-449bff7060b4/Work/logsagenttechnicaldepth/fit.p
y:144: UserWarning: Using a target size (torch.Size([16, 112, 112])) that is
different to the input size (torch.Size([16, 1, 112, 112])) is deprecated. Pl
ease ensure they have the same size.
    loss = crit(out,data)
100%|██████████| 1202/1204 [01:56<00:00, 7.77it/s]/mnt/5e85af0b-fa6b-4dca-a5
39-449bff7060b4/Work/logsagenttechnicaldepth/fit.py:144: UserWarning: Using a
target size (torch.Size([4, 112, 112])) that is different to the input size
(torch.Size([4, 1, 112, 112])) is deprecated. Please ensure they have the sam
e size.
    loss = crit(out,data)
```

```
In [54]: anml_rec = torch.tensor(test_item.test_loss)
np.save(path+'anml_rec.npy',anml_rec_gen)
```

```
In [61]: fig , ax = plt.subplots(1,2,figsize=(18,4))
ax[0].bar(['val_disc', 'test_disc','anml_rec'],[np.median(val),np.median(anm
l_eth_short),np.median(anml_rec)])
ax[0].set_title(f' Medians')
ax[0].set_ylabel('Binary cross entropy')

ax[1].boxplot([val,list(anml_eth_short),list(anml_rec)],showfliers = False);
ax[1].set_xticklabels(['Val','Test','anml_rec']);
ax[1].set_title(f' distribution')
ax[1].set_ylabel('Binary cross entropy')
```

```
Out[61]: Text(0, 0.5, 'Binary cross entropy')
```



Very Good result approaches the criteria we are Looking for